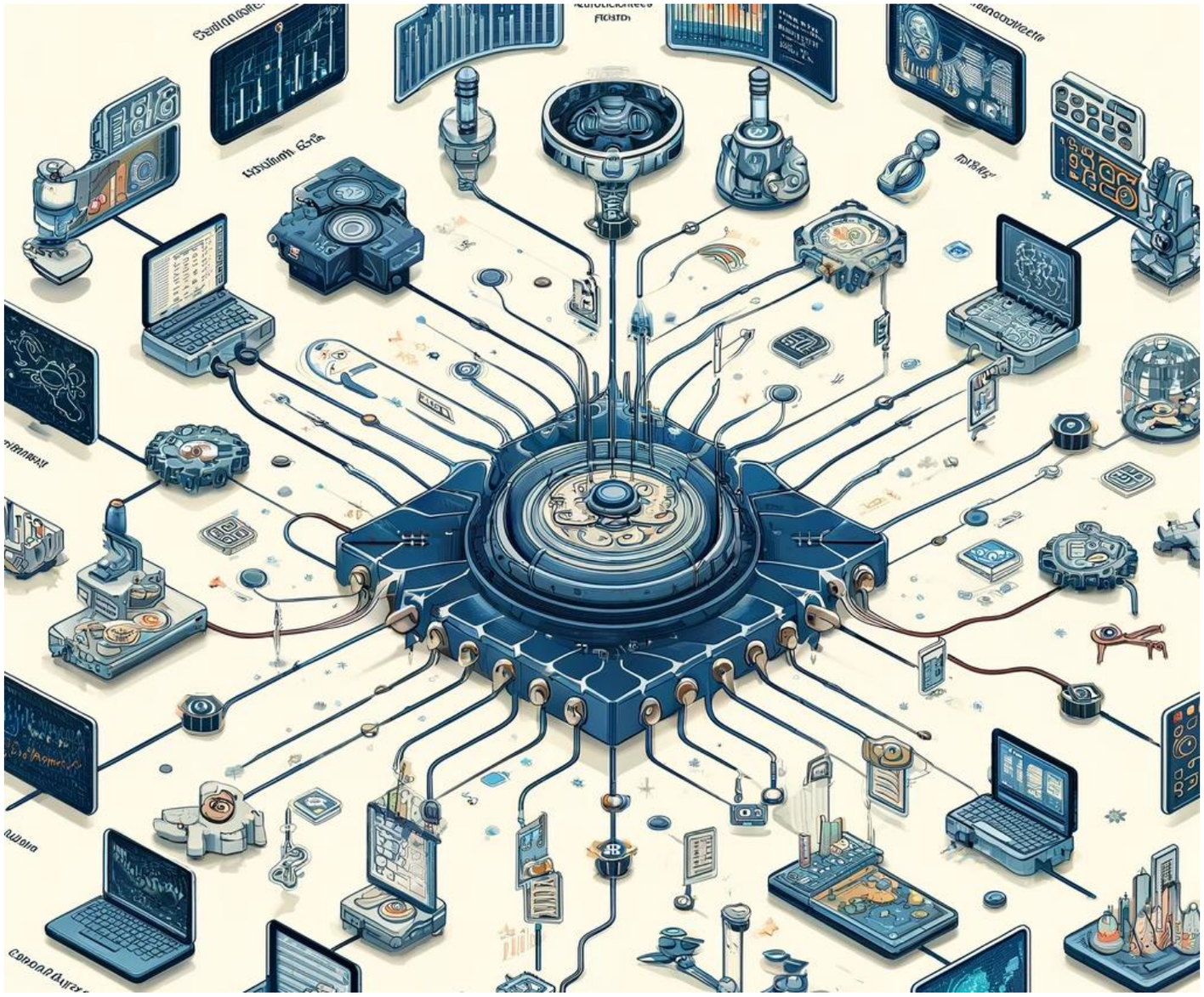
Exploring Scientific Workflows with CWL and dispel4py

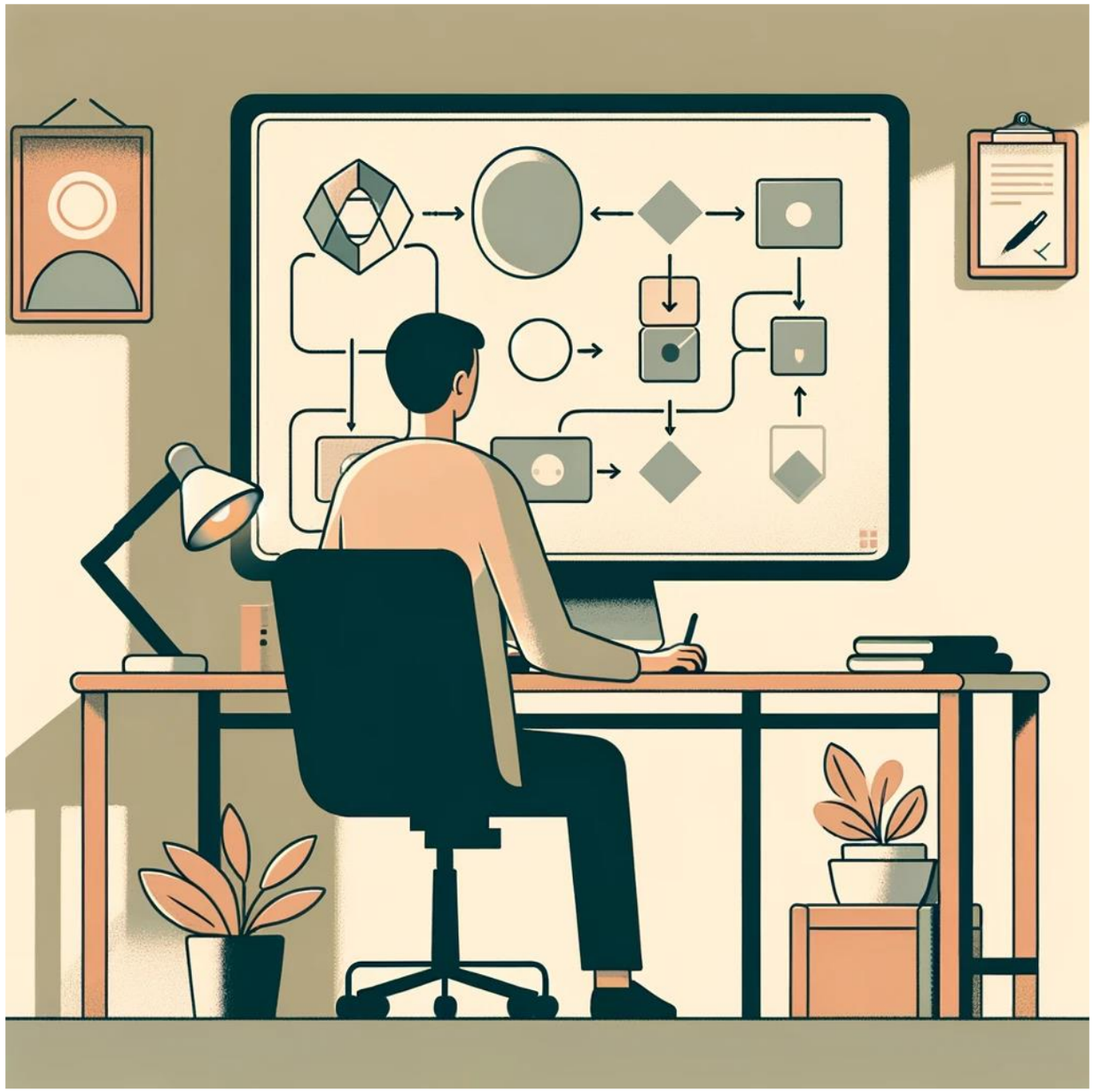
Day2 -Module 4

- Dr. Rosa Filgueira
- Chancellor's Fellow AI & DS
- University of Edinburgh, EPCC
- r.filgueira@epcc.ed.ac.uk



dispel4py Advanced Concepts

- Inputs and outputs
- Graph examples
- Composite PEs
- Chains
- Groupings
- Mappings to execution platforms
- Running graphs
- Useful dispel4py Information
- RA example
- Provenance

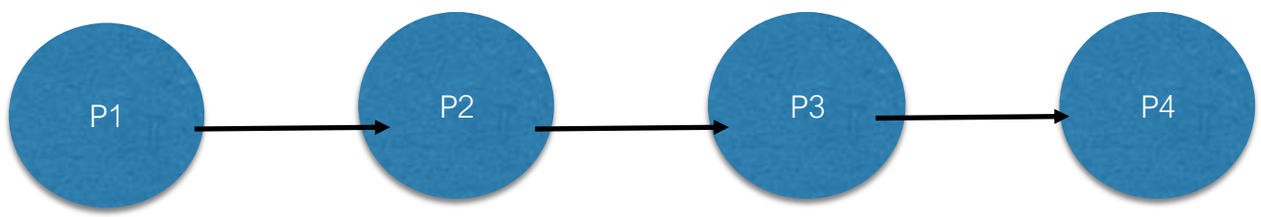


Inputs and outputs

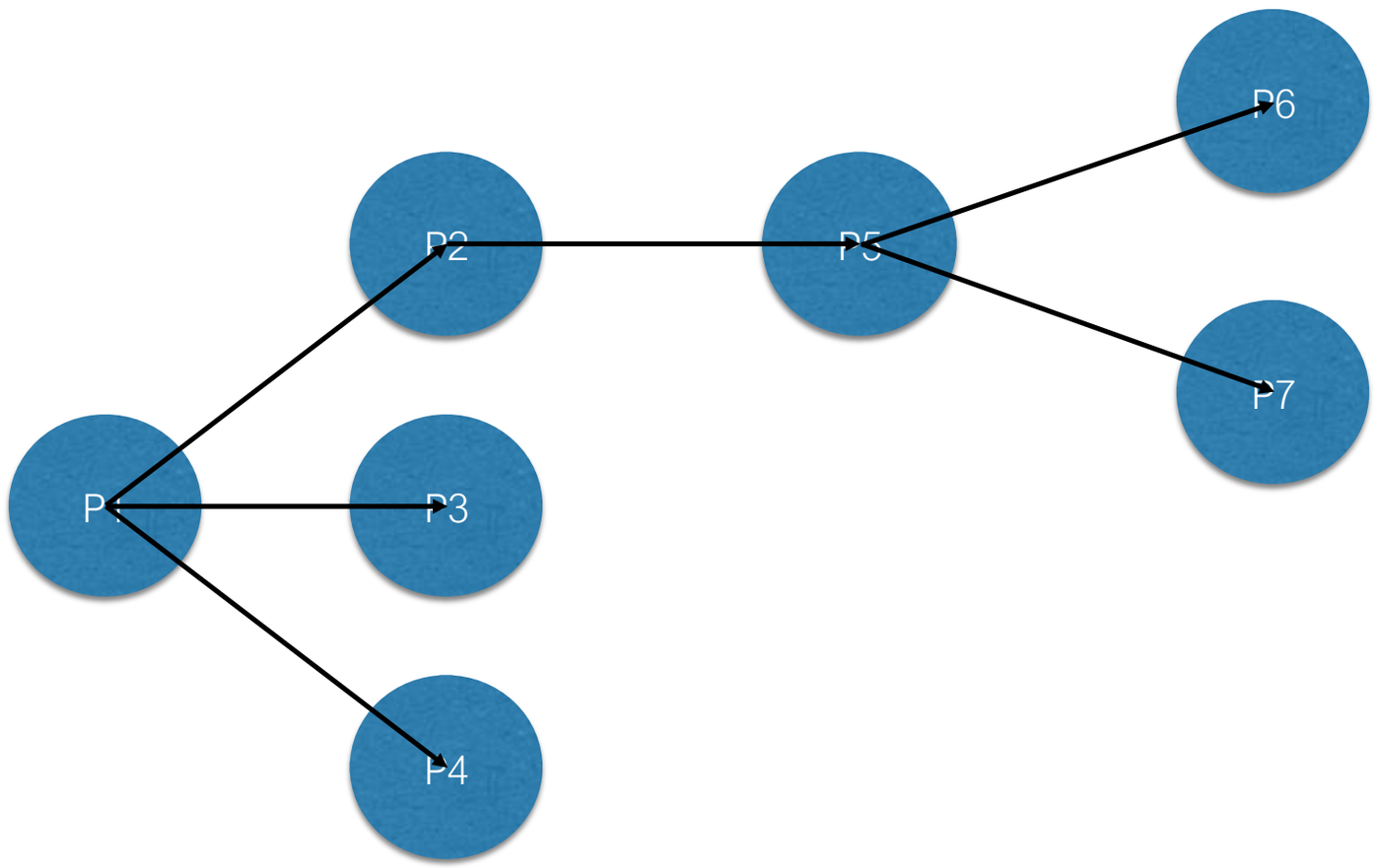
- How many pieces of data does the PE combine?
  - Transformation: one input (e.g. normalisation)
  - Product or Join: two inputs (e.g. cross-correlation)
  - And many others!

- What is the rate of processing?
  - One output per input (transformation, filter)
  - Aggregation of data (e.g. stacking)

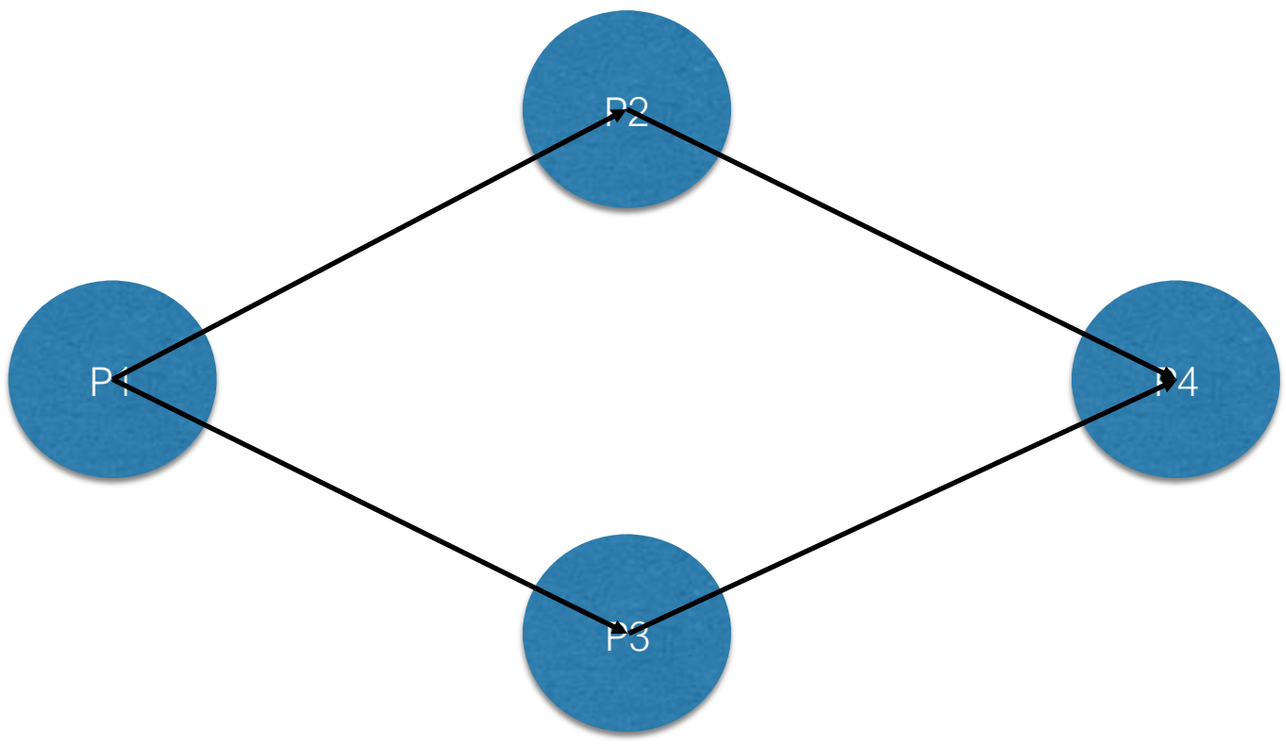
Pipeline



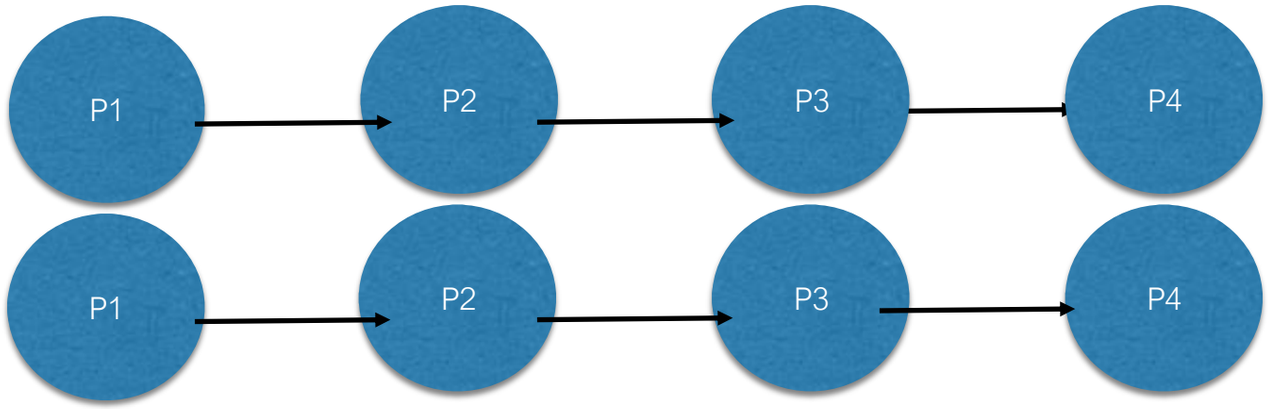
Tree



Split and Merge

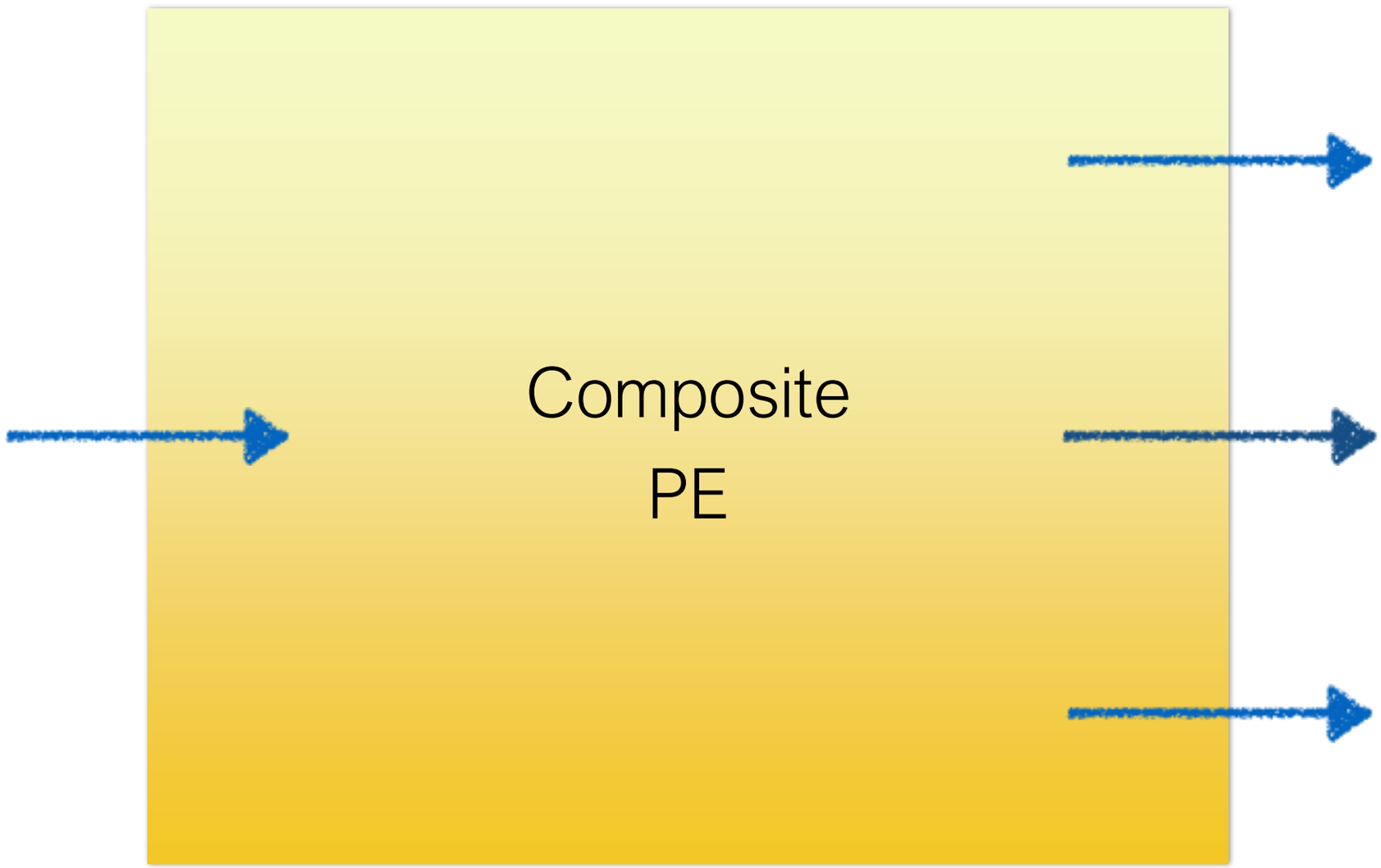


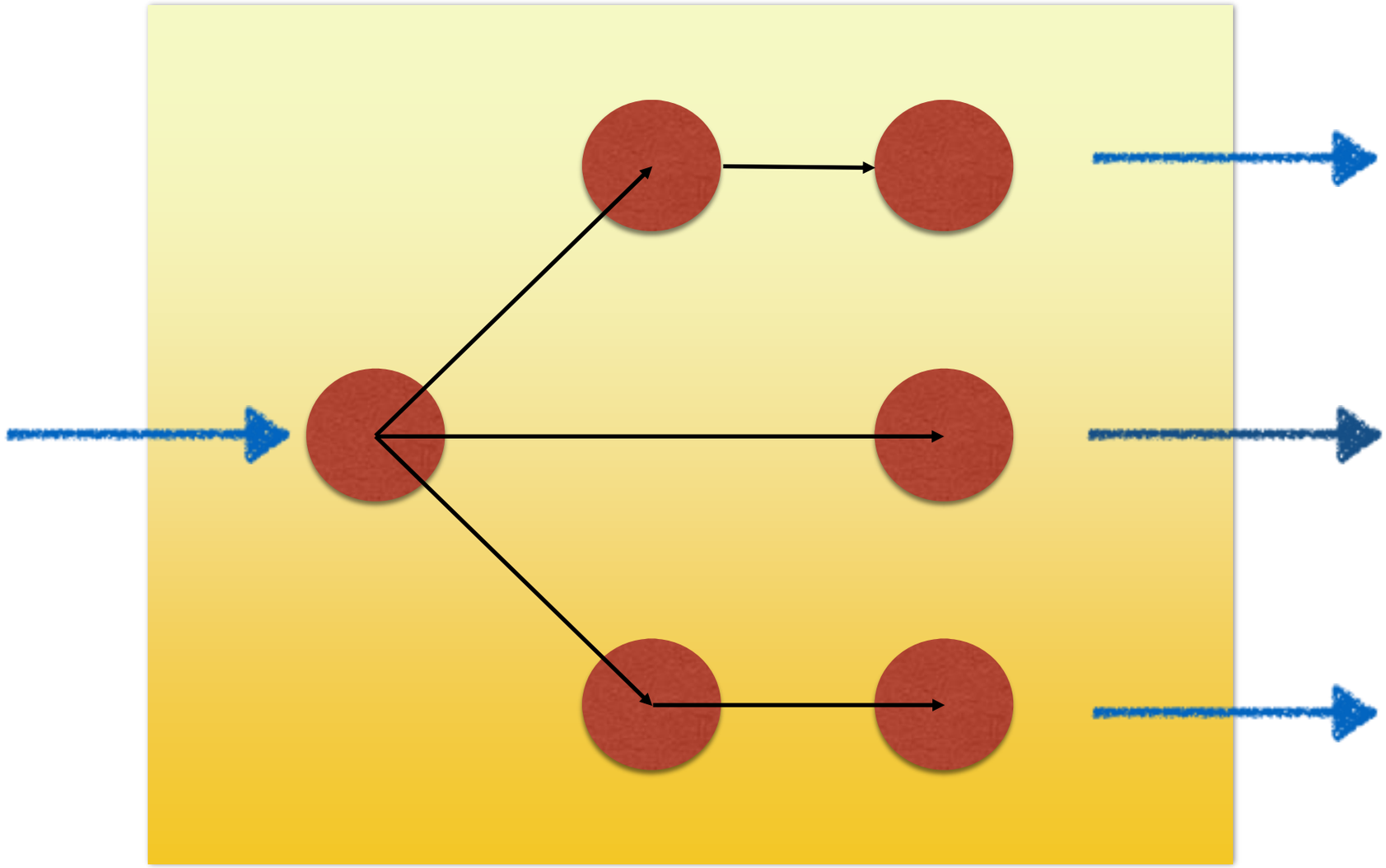
Unconnected

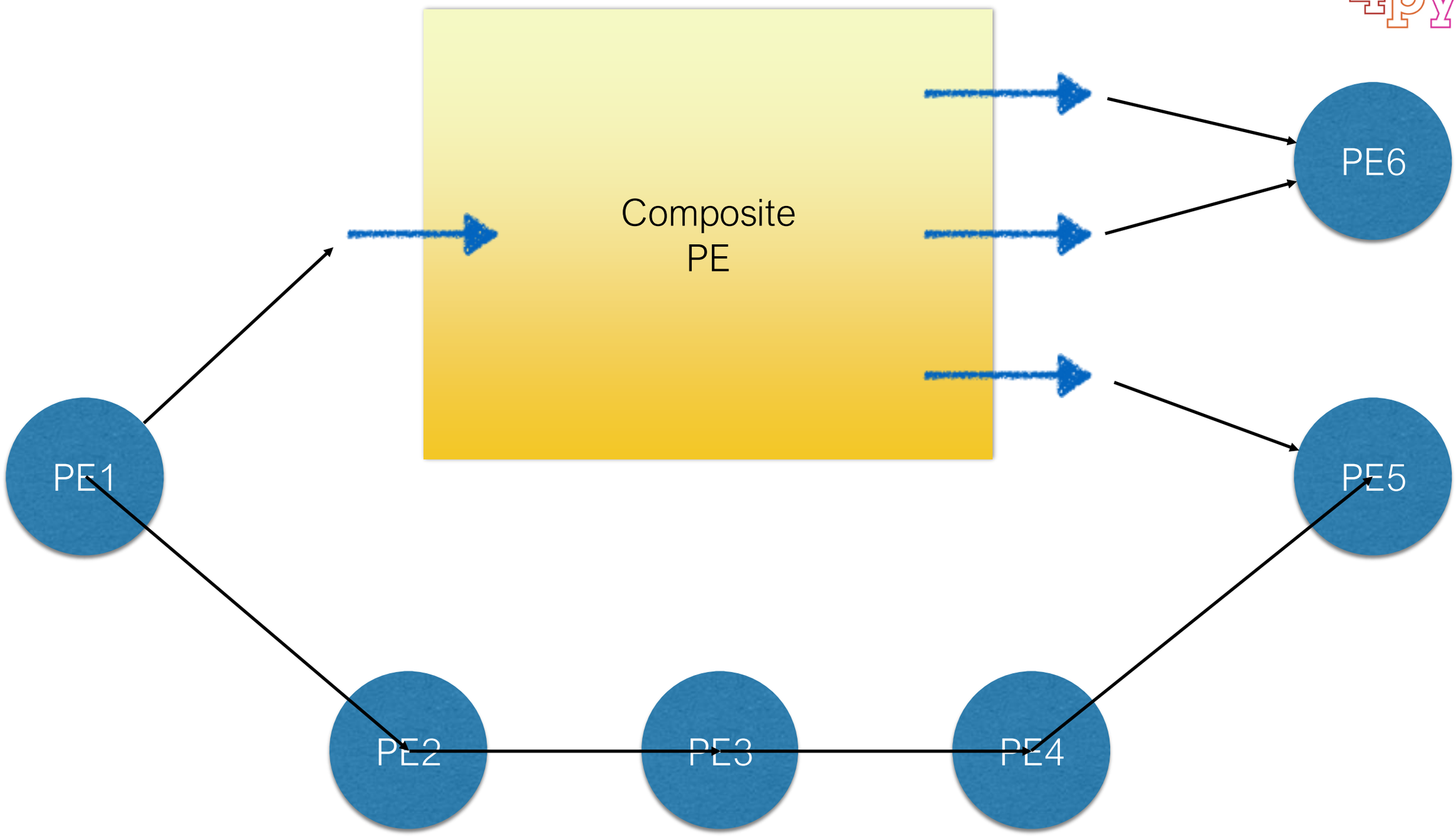


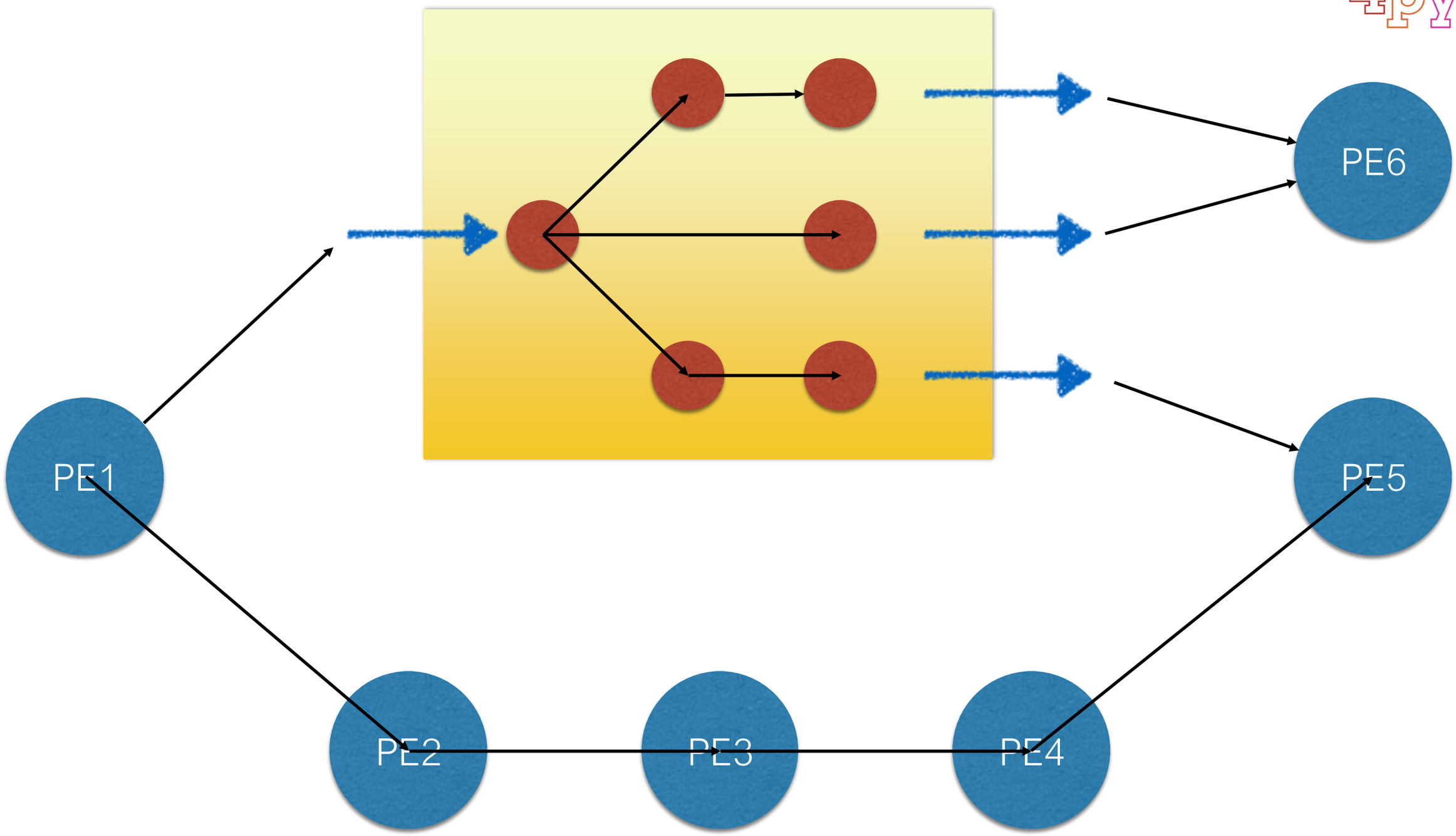
Composite PEs

- A composite PE is a nested graph
- Looks like a PE but contains other PEs
- Hides the complexity of an underlying process
- When creating a graph, a composite PE is treated like any other PE


Composite
PE




Composite
PE
PE1
PE2
PE3
PE4
PE5
PE6

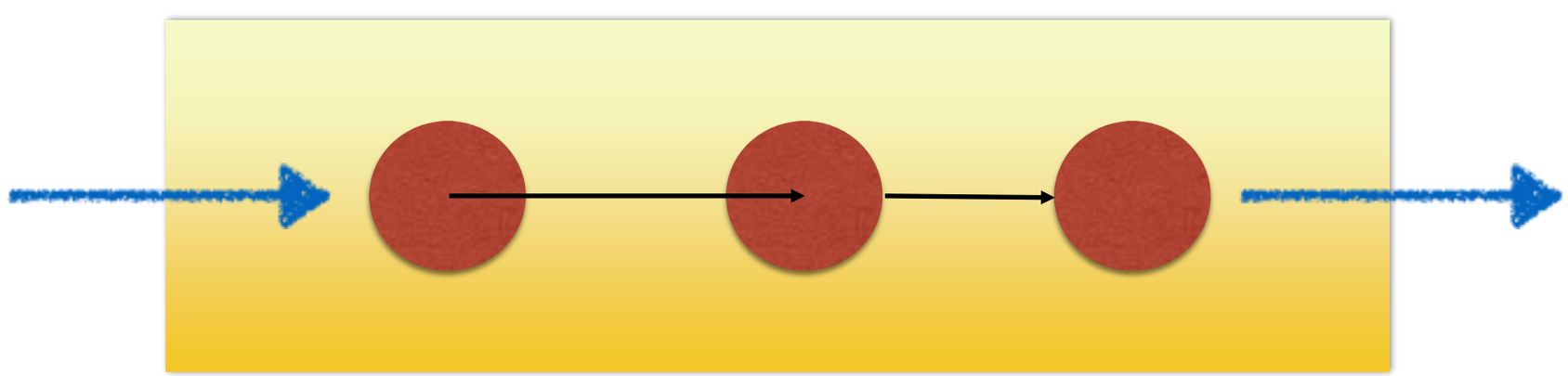

PE1
PE2
PE3
PE4
PE5
PE6

Why composite PEs?

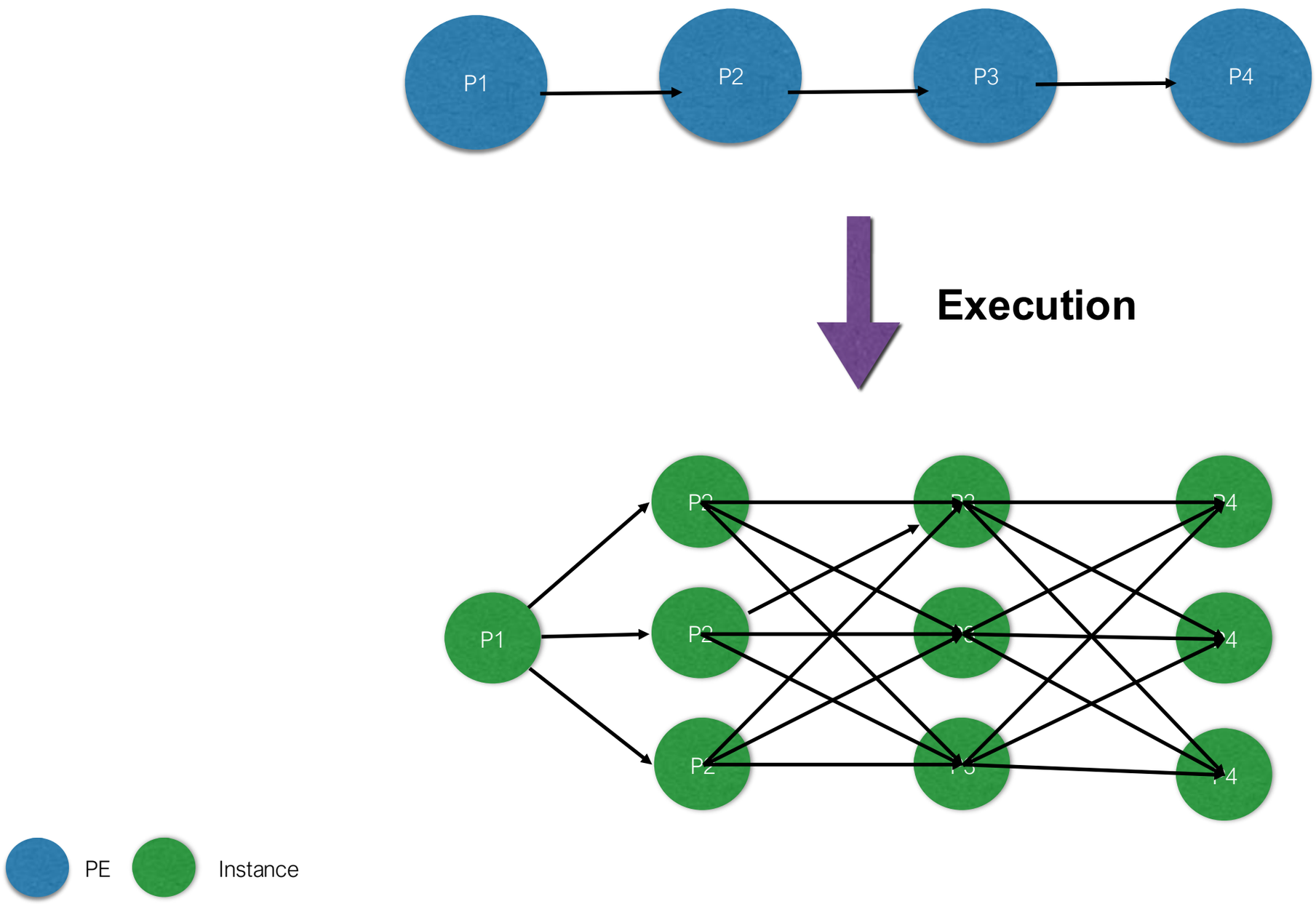
dispel4py provides a utility to create a composite PE with a chain of processing PEs:

1. Only implement the processing functions (process one, return one)
2. Create a list that contains the functions in the order that they are to be applied
3. Pass this list to the utility and receive a composite PE in return

```
def decimate(data, sps):
    st = data[0]
    st.decimate(int(st[0].stats.sampling_rate/sps))
    return st
def detrend(data):
    st = data[0]
    st.detrend('simple')
    return st
def demean(data):
    st = data[0]
    st.detrend('demean')
    return st

compositePE = create_iterative_chain([(decimate, {'sps':4}), detrend, demean])
```



Executing graphs

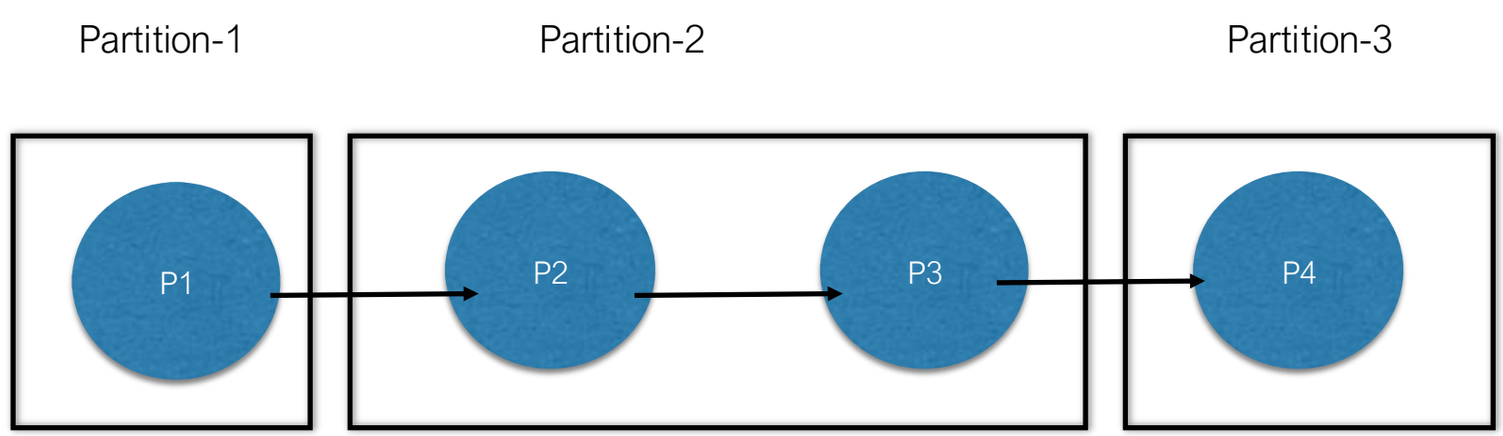


4 PEs

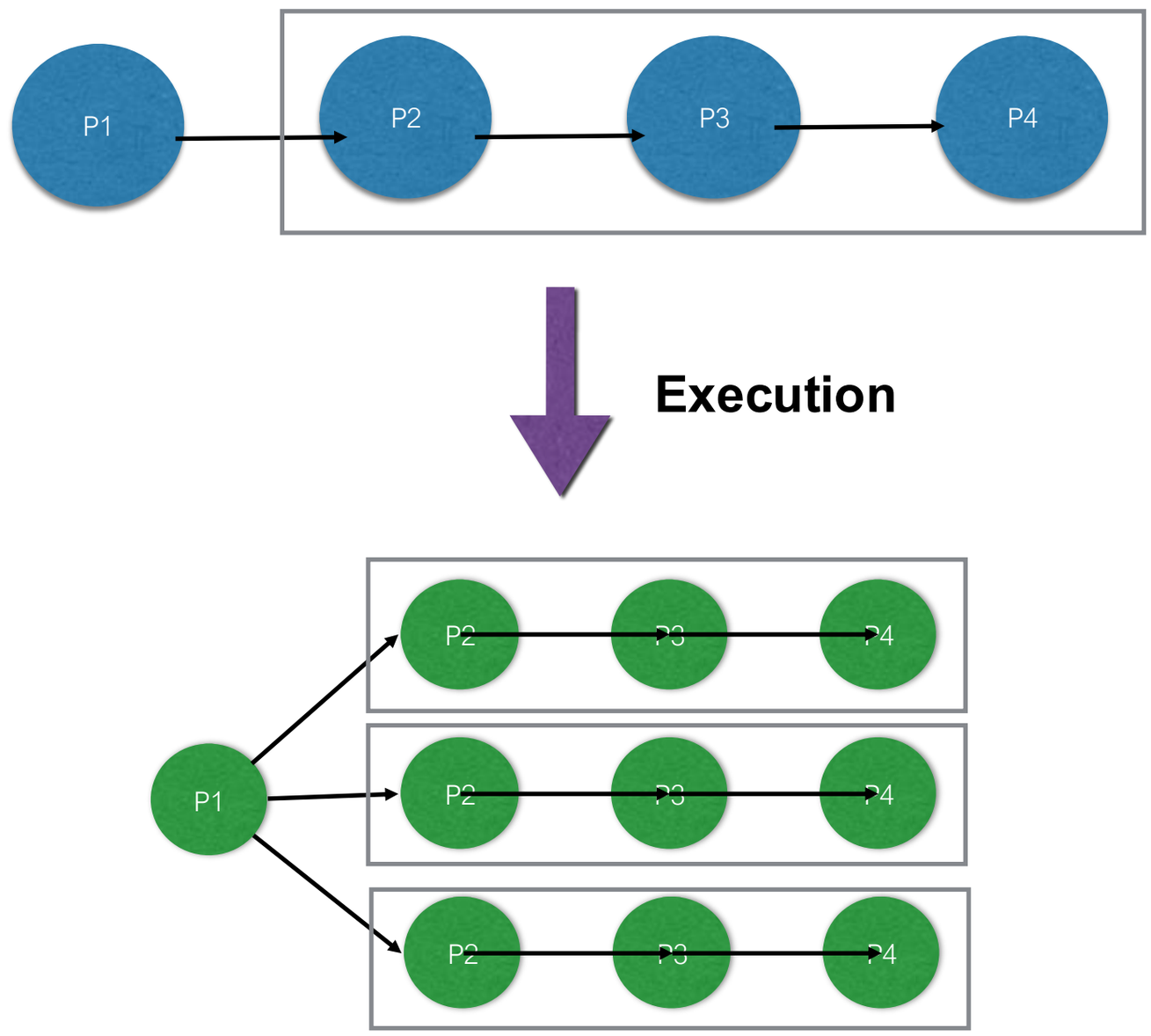
10 processes/instances

Partitions

- Run several PEs in a single process
- User defined
- Applies to parallel environments



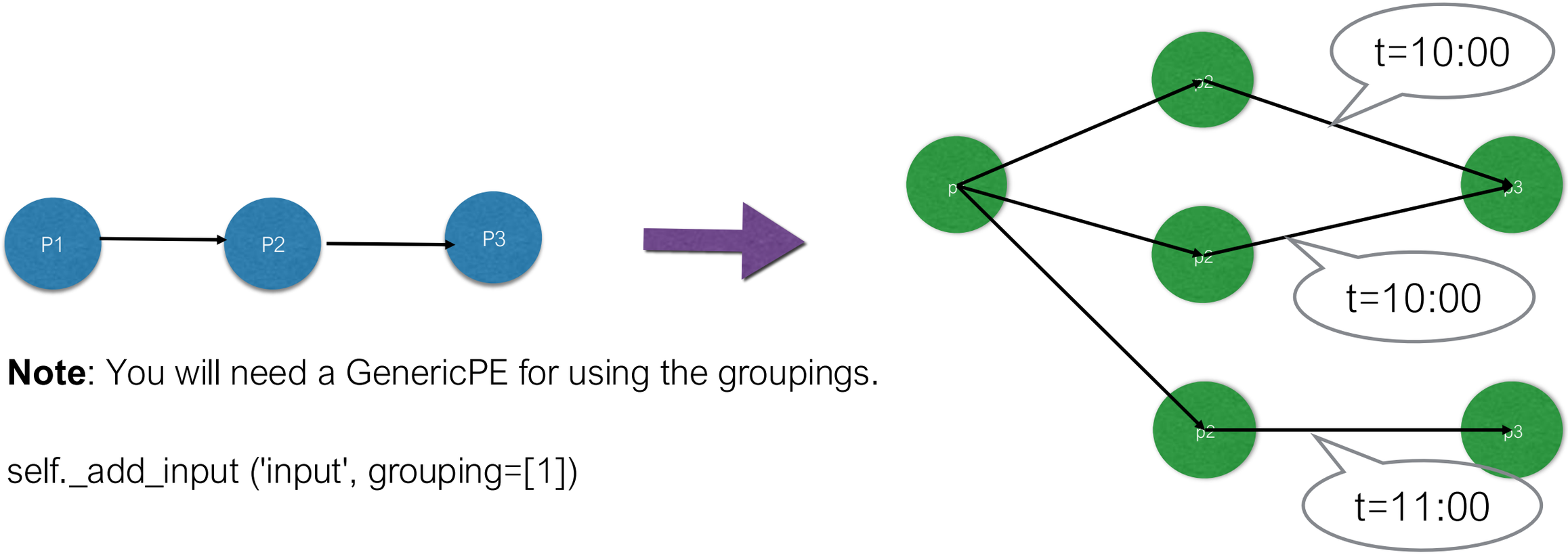
Partitioned Pipeline



Groupings

“Grouping by” a feature (MapReduce)

All data items that satisfy the same feature are guaranteed to be delivered to the same **instance** of a PE



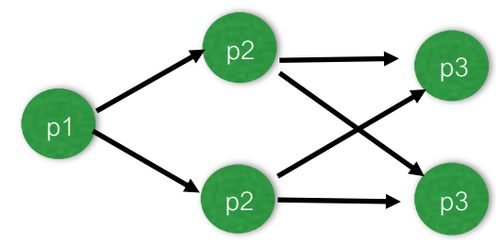
Note: You will need a GenericPE for using the groupings.

self._add_input ('input', grouping=[1])

Other groupings

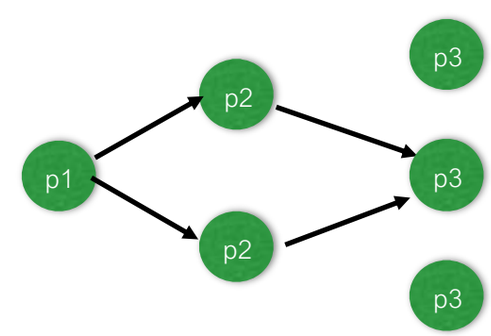
One-To-All

P3 - grouping "all":
P2 instances send copies of their output data to **all** the connected instances



Global

P3 - grouping "global":
All the instances of P2 send all the data to **one** instance of P3



Mappings

Simple process

- Sequential mapping for local testing

Multi process

- Parallelism based on processes, using Python's multiprocessing library
- Shared memory

MPI

- Distributed Memory, message-passing parallel programming model
- Practical, portable, efficient, flexible and stable
- Many HPC centres support it, and it has been widely used in the HPC community

Note: Now we have also REDIS and Dynamic Mappings

Running graphs

Sequential mapping

>> dispel4py simple <name_dispy_graph> <-f input_file in JSON format>

E.g: dispel4py simple dispel4py.examples.graph_testing.pipeline_test

Multi-process mapping

>> dispel4py multi <name_dispy_graph> -n <number mpi_processes> <-f input_file in JSON format> <-s>

E.g : dispel4py multi dispel4py.examples.graph_testing.pipeline_test -n 6

MPI mapping

>> mpiexec -n <number mpi_processes> dispel4py mpi <name_dispy_graph> <-f input_file in JSON format> <-s>

E.g : mpiexec -n 6 dispel4py mpi dispel4py.examples.graph_testing.pipeline_test

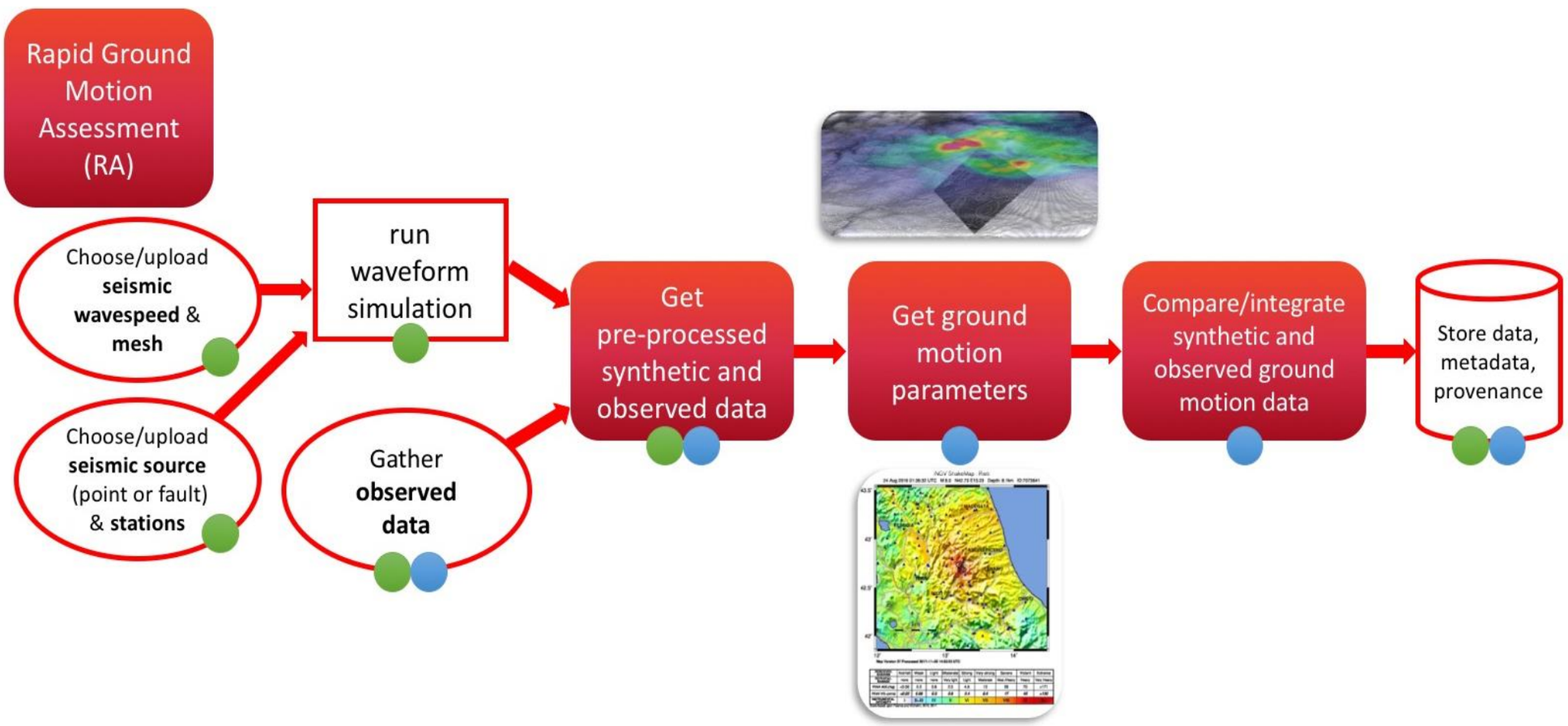
Useful dispel4py information

- dispel4py commands:
  - -h, --help >> show this help message and exit
  - -a attribute, --attr attribute >> name of graph variable in the module
  - -f inputfile, --file inputfile >> file containing input dataset in JSON format
  - -d inputdata, --data inputdata >> input dataset in JSON format
  - -i iterations, --iter iterations >> number of iterations

Useful dispel4py information

- inputs to the workflow:
  - dispel4py simple workflow.py -d ‘{“PE NAME CLASS" : [{"input" : “Hello World World”}]}’
  - dispel4py simple workflow.py -d ‘{"PE NAME CLASS" : [{"input" : "coordinates.txt"}]}’
  - dispel4py simple workflow.py -f coordinates.txt

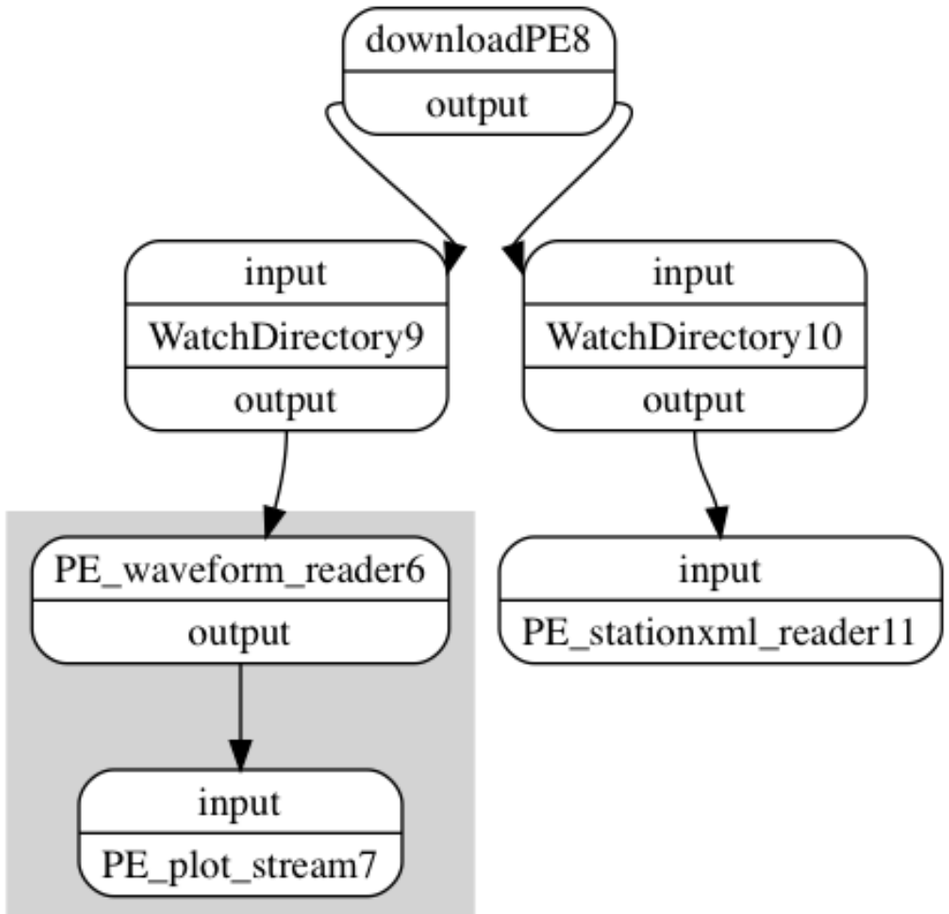
Rapid Ground Motion Assessment

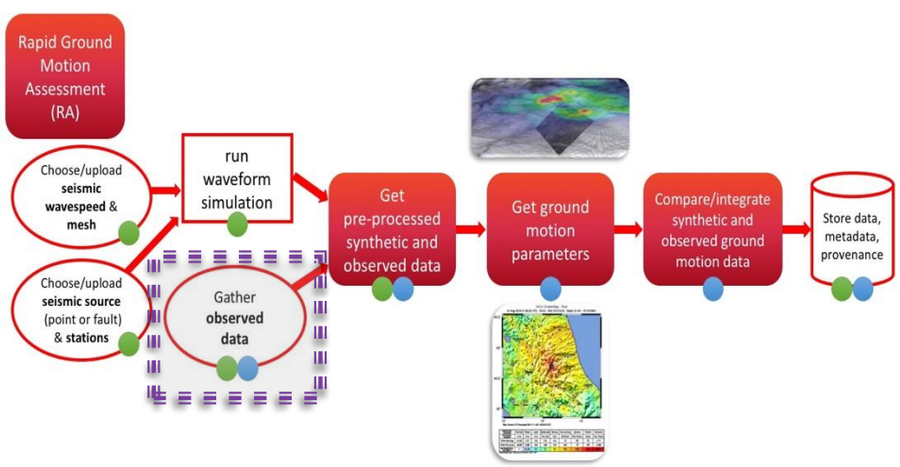


Computing Needed

Validated trace

RA – Download observed data

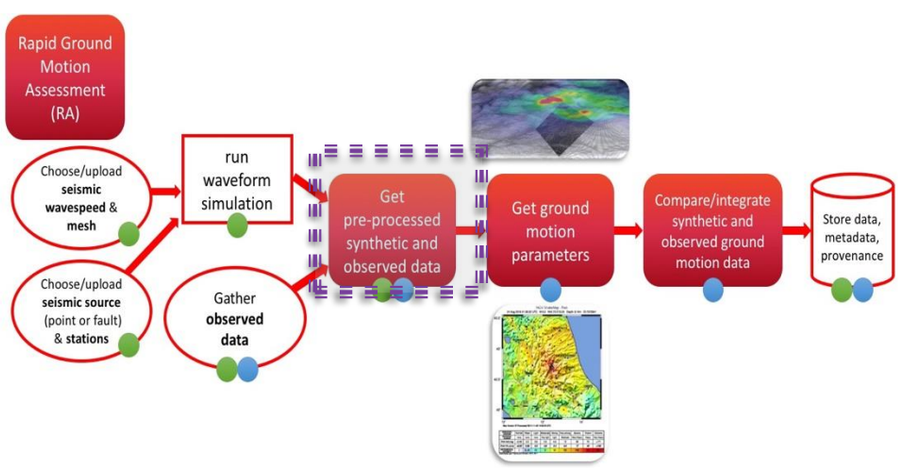


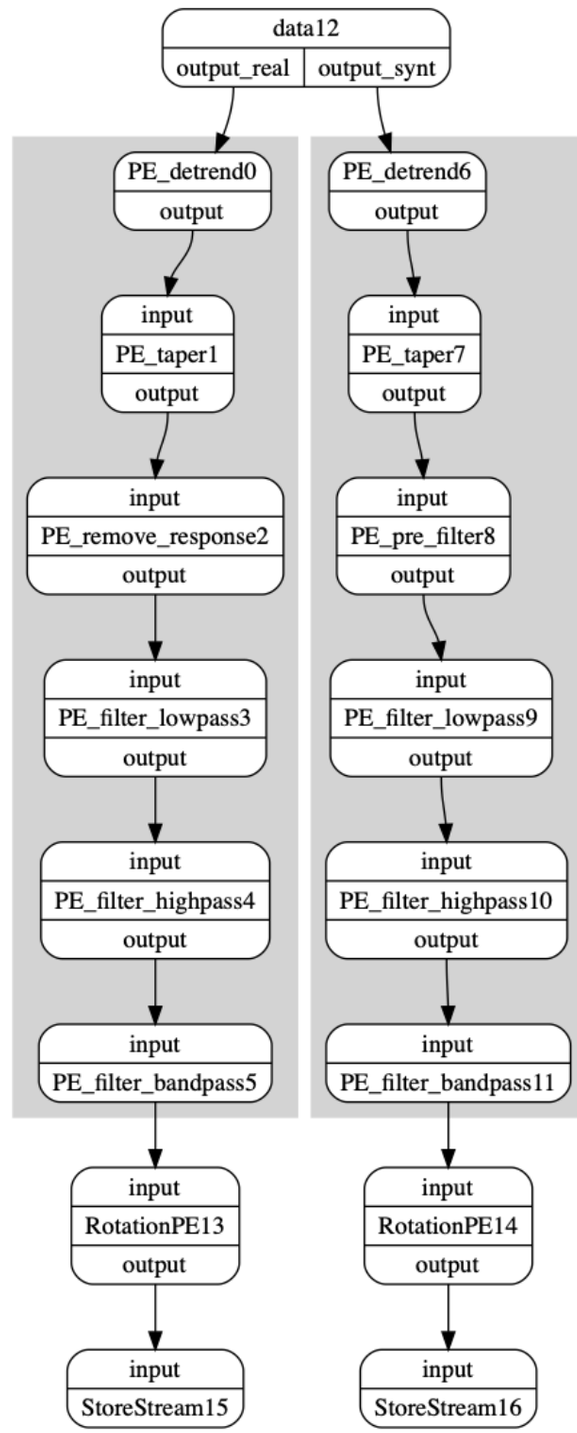


The workflow downloads real waveforms corresponding to the same earthquake.


```
downloadPE:
  0:
    input:
      minimum_interstation_distance_in_m:   100
      channel_priorities:
        0:                                  "BH[E,N,Z]"
        1:                                  "EH[E,N,Z]"
      location_priorities:
        0:                                  ""
        1:                                  "00"
        2:                                  "10"
    mseed_path:                             "./data"
    stationxml_path:                        "./stations"
    RECORD_LENGTH_IN_MINUTES:               1
    ORIGIN_TIME:                            "2013-02-16T21:16:09.29"
    minlatitude:                            41.10007459633125
    maxlatitude:                            42.89777970948071
    minlongitude:                           12.041644551237324
    maxlongitude:                           14.439665626790928
```

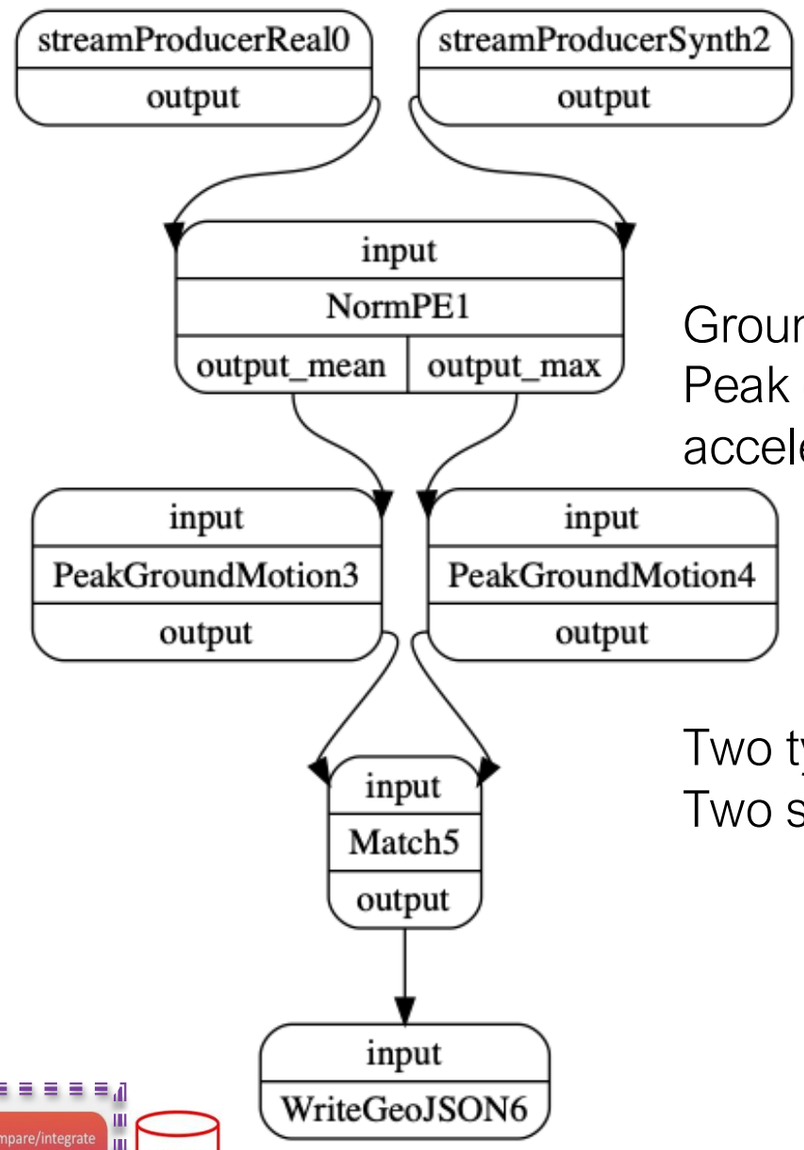
RA – Preprocessing observed and synthetic





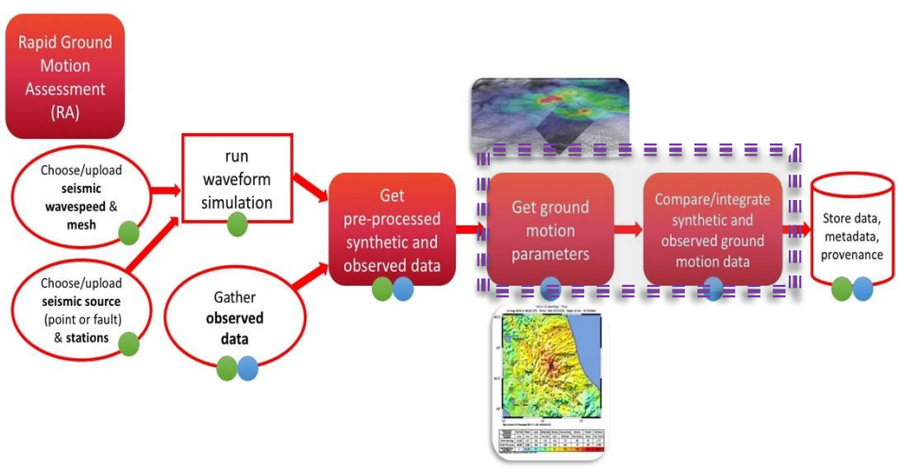
Similar preprocessing steps in synthetic and observed data

RA – Ground motion parameters



Ground motion parameters:
Peak ground values of displacement, velocity and acceleration.

Two types of normalisation – Mean & Max
Two set of PGM outputs



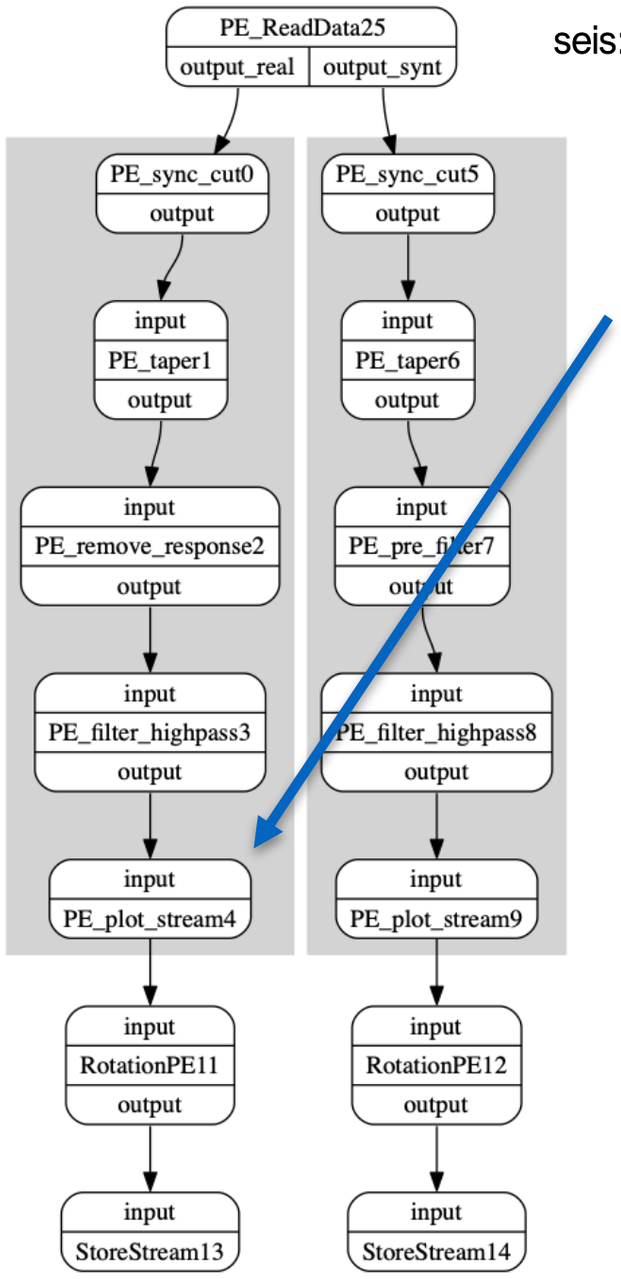
Provenance

Inline metadata injection

Waveform Preprocessing

pipeline JSON Description (eg. from file)

Manual Extensions



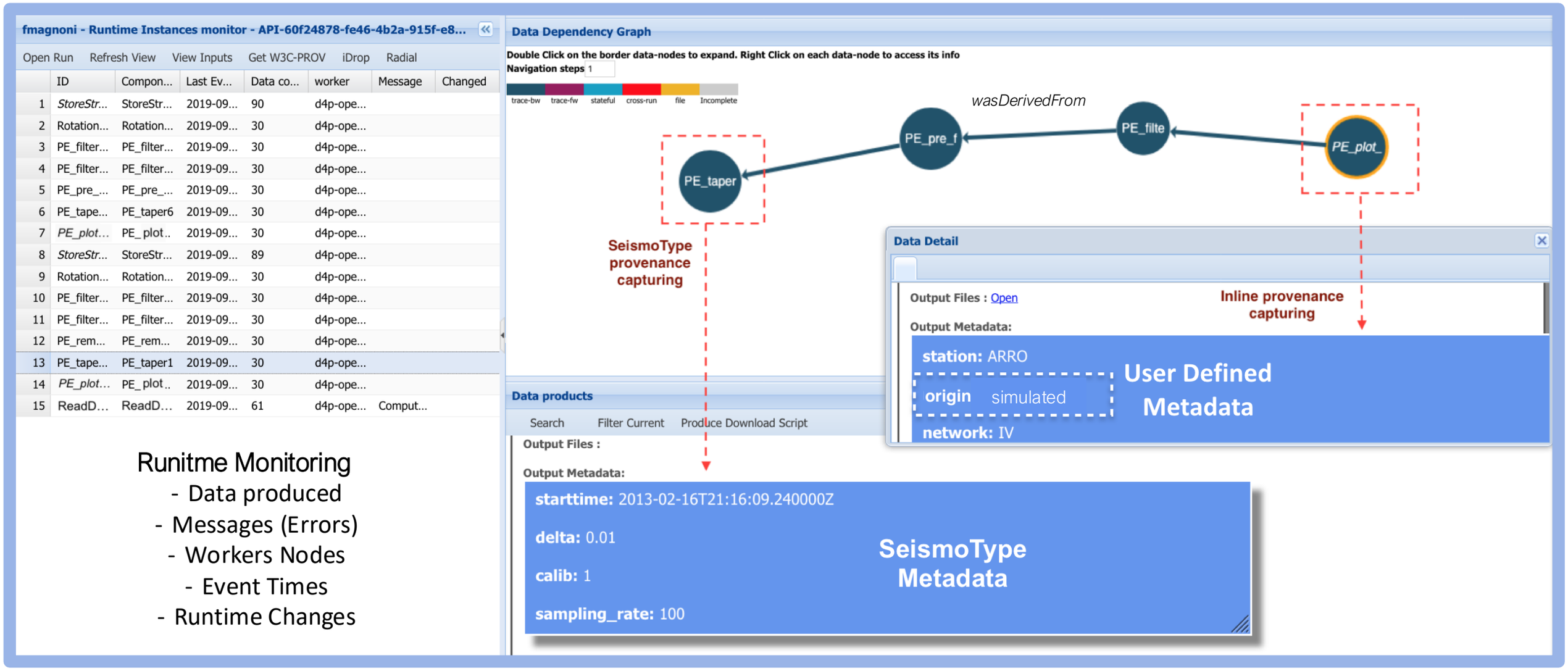
seis:preprocessing

Functions encoded in Python
User Defined Metadata injection into Lineage traces

```
def plot_stream(stream, output_dir, tag):
    stats = stream[0].stats
    filename = "%s.%s.%s.%s.png" %
        (stats['network'], stats['station'],
         stats['channel'], tag)

    path = os.environ['STAGED_DATA'] + '/' + output_dir
    dest = os.path.join(path, filename)
    stream.plot(outfile=dest)

    prov = {'location': "file://" + socket.gethostname()+"/"+dest,
            'format': 'image/png',
            'metadata': {'origin': tag}}

    return {'_d4p_prov': prov, '_d4p_data': stream}
```

User Defined Metadata

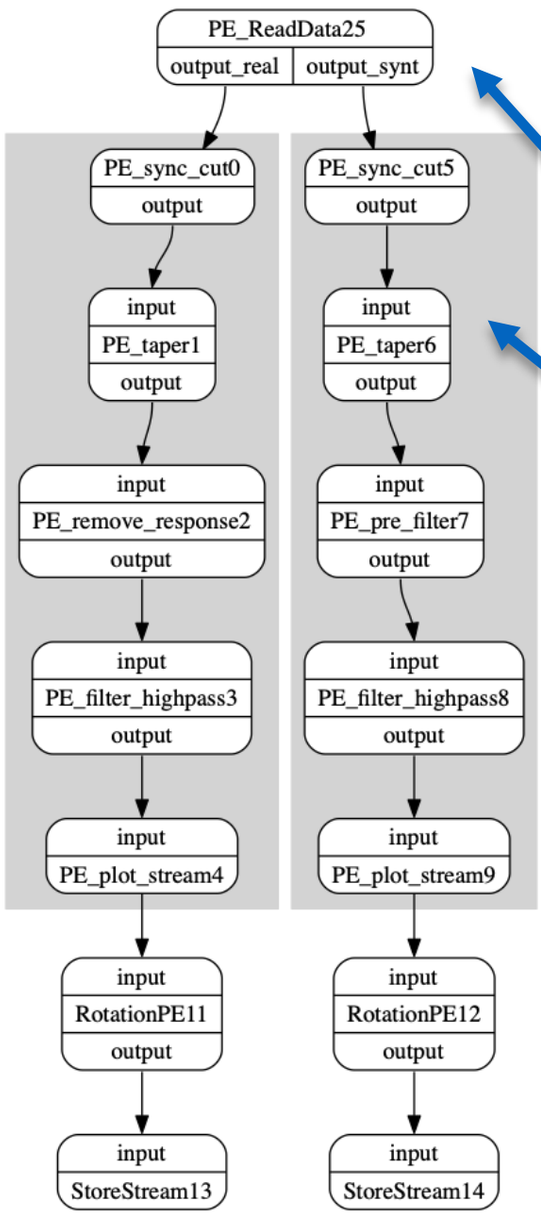
Monitor, search and analyse results through lineage



S-ProvFlow:https://gitlab.com/project-dare/s-Pro

Workflow Contextualisation

Seismic Data Preprocessing



Configuration Profile in JSON with Provenance Types

```
{
    'provone:User': "aspinuso",
    's-prov:description' : "provdemo",
    's-prov:workflowName': "waveform preprocessing pipeline",
    's-prov:workflowType': seis:preprocessing",
    's-prov:WFExecutionInputs' :  [{...}],
    's-prov:save-mode'    : 'service',
    's-prov:WFExecutionInputs':  [{...}],
    # defines the Provenance Types and Provenance Clusters for the Workflow's Components
    's-prov:componentsType' :
            {'s-prov:componentsType' :
                {'PE_ReadData':       { 's-prov:type':['SeismoType'],
                                        's-prov:prov-cluster': seis:DataHandler',

                'PE_taper':           { 's-prov:type':['SeismoType'],
                                        's-prov:prov-cluster': seis:Processor',

                'PE_remove_response':{ 's-prov:type':['SeismoType'],
                                        's-prov:prov-cluster': seis:Processor'},

                'PE_plot_stream':     {'s-prov:type'
```

Semantic Tagging

starttime: 2013-02-16T21:16:09.240000Z

delta: 0.01

calib: 1

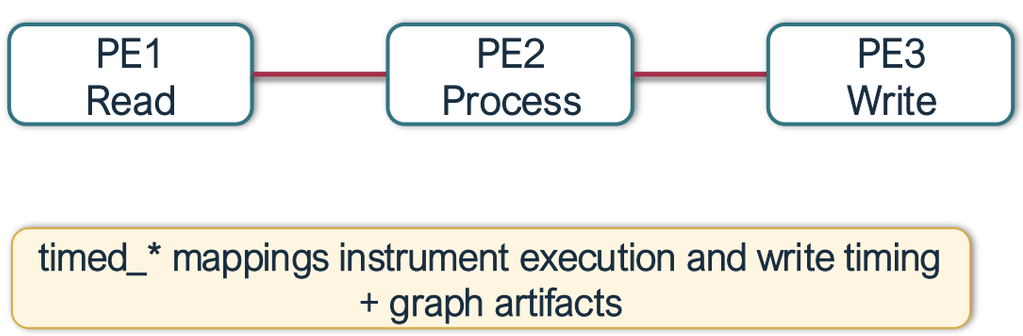
sampling_rate: 100

ProvenanceType for Metadata Contextualisation

Spinuso, A, Atkinson, M & Magnoni, *Active provenance for Data-Intensive workflows: engaging users and developers.*

IEEE eScience 2019 proceedings. Bridging from Concepts to Data and Computation for eScience (BC2DC'19) Workshop

Monitoring dispel4py workflows



Why monitor?

Check where time is spent in a dataflow graph

Compare sequential, local parallel, and MPI runs

Detect load imbalance between PE instances/ranks

Explain the runtime graph created by a mapping

Monitoring dispel4py workflows

Timed Mappings

timed_simple
Single process
Best for logic checks, reproducibility, and baseline timing.

timed_multi
Local multiprocessing
Best for studying local parallelism and PE instance distribution.

timed_mpi
MPI / distributed ranks
Best for HPC or cluster-style performance analysis.

Commands pattern

```
dispel4py timed_simple <workflow_module> -i 10 --print-shape
dispel4py timed_multi  <workflow_module> -i 10 -n 10 --print-shape
mpiexec -n 10 dispel4py timed_mpi <workflow_module> -i 10 --print-shape
```


https://github.com/StreamingFlow/d4py/blob/main/README_Monitor.md

Monitoring dispel4py workflows

Shared flags across the *Timed Mappings*

- `--timing-dir` : output directory (default: `timings`)
- `--timing-prefix` : filename prefix (default: `monitor`)
- `--run-id` : custom run identifier; if omitted, generated automatically
- `--summary-file` : output path for per-PE summary CSV
- `--instance-summary-file` : output path for per-instance summary CSV
- `--iteration-summary-file` : output path for merged per-iteration CSV
- `--iteration-latency-summary-file` : output path for per-instance latency stats derived from iteration timings
- `--shape-file` : output path for abstract graph JSON
- `--concrete-shape-file` : output path for concrete graph JSON
- `--abstract-figure-file` : output path for abstract graph PNG
- `--concrete-figure-file` : output path for concrete graph PNG
- `--no-graph-figures` : skip PNG generation
- `--print-shape` : print abstract and concrete shape details to stdout

./timings/

You may see files such as:

- `monitor_shape_run<id>.json`
- `monitor_concrete_shape_run<id>.json`
- `monitor_summary_run<id>.csv`
- `monitor_instances_run<id>.csv`
- `monitor_iteration_timings_run<id>.csv`
- `monitor_iteration_timings_summary_run<id>.csv`
- `monitor_<PE>_rank<R>_run<id>.csv`
- `monitor_iterations_<PE>_rank<R>_run<id>.csv`
- `monitor_abstract_graph_run<id>.png`
- `monitor_concrete_graph_run<id>.png`

Monitoring dispel4py workflows

Shared flags across the *Timed Mappings*

Every timed mapping writes files to a monitoring directory, by default ./timings.

./timings/

Timing CSVs

Per-PE summaries

Per-instance summaries

Per-iteration traces

Graph JSON

Abstract graph

Concrete runtime graph

Process allocation

Latency stats

min, p50, p95, max

Avg and totals

Graph PNGs

Abstract workflow figure

Concrete instance figure

Monitoring dispel4py workflows

Start coarse, then drill down to individual PE instances and iterations.

1. PE summary (PE level) — Which PE is the hotspot?
2. Instance summary (Instance level) — Are ranks balanced?
3. Iteration traces (Iteration level) — Are there slow/outlier items?

```
PE level: WordCounter
Instance level: WordCounter@1, WordCounter@2, WordCounter@3
Iteration level: each time WordCounter@1/2/3 processes one item
```

`avg_secs` Mean processing time; useful, but can hide outliers.

`p50_secs` Median latency: half of iterations were faster, half slower.

`p95_secs` Tail latency: highlights slower behaviour for most runs.

`max_secs` Slowest observed iteration; useful for rare extreme delays.

Rule of thumb

If p95 is much larger than p50, the workflow has jitter or outliers. If one rank has much higher total time, check load balance.

dispel4py Advanced Concepts

Hands-On Exercises:

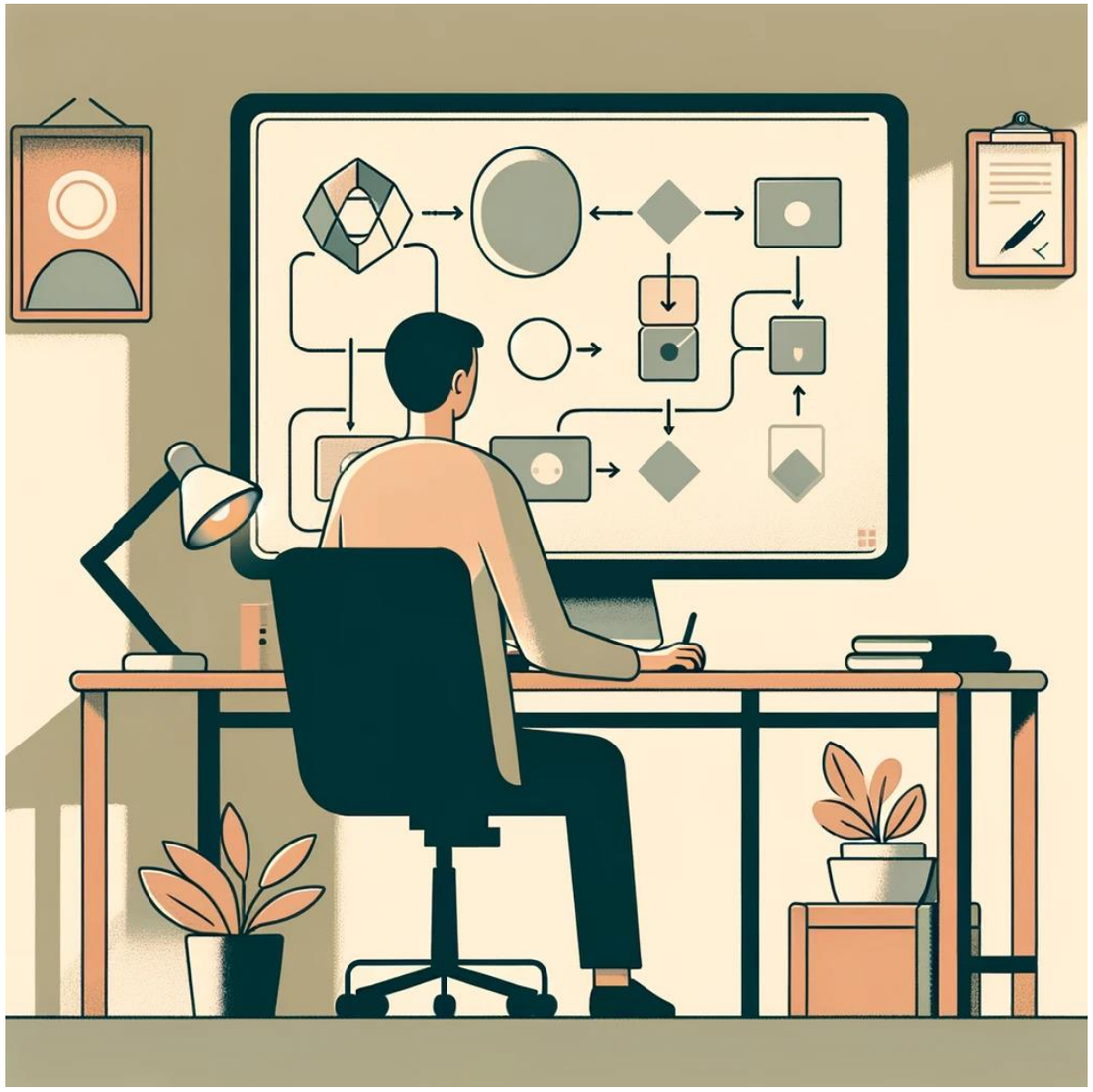
- Intro_Tutorial_dispel4py_2_0.ipynb
- Testing_dispel4py_2_0.ipynb

Additional Material (not covered in the seminar)

Complex Scientific Use Cases:
https://github.com/StreamingFlow/d4py_workflows

Monitoring dispel4py workflows:

Monitoring_Traces_dispel4py_2_0.ipynb



dispel4py & CWL

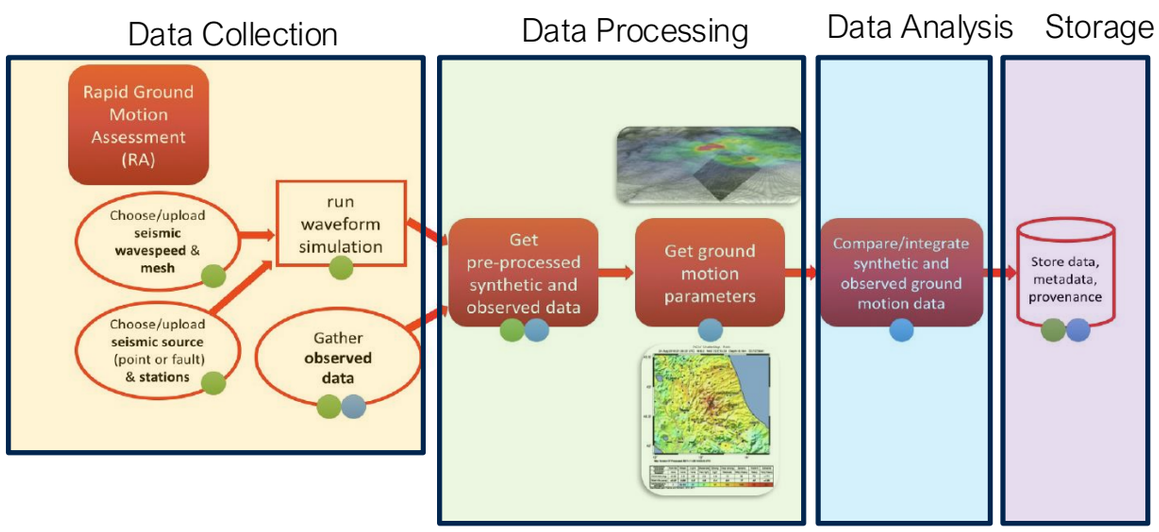
- Using dispel4py for implementing the RA subcomponents of the application
- Using cwl for connecting the different dispel4y RA subcomponents

ra.cwl

```
#!/usr/bin/env cwl-runner

cwlVersion: v1.0
class: Workflow

inputs:
   create_env_script: File
   download_workflow: File
   download_argument_f: File
   preprocess_workflow: File
   ra_workflow: File
   ra_argument_d: string
outputs:
    misfit_data:
        type: Directory
        outputSource: preprocess_data/output
    pgm_data:
        type: Directory
        outputSource: rapid_assessment/output
steps:
  create_env:
    run: create_env.cwl
    in:
      script: create_env_script
    out: [output]
  download_data:
    run: dispel4py_download.cwl
    in:
      workflow:  download_workflow
      argument_f: download_argument_f
      misfit_data: create_env/output
    out: [output]
  preprocess_data:
    run: dispel4py_preprocess.cwl
    in:
      workflow: preprocess_workflow
      misfit_data: download_data/output
    out: [output]
  rapid_assessment:
    run: dispel4py-RA-pgm_story.cwl
    in:
      workflow: ra_workflow
      argument_d: ra_argument_d
      misfit_data: preprocess_data/output
    out: [output]
```

ra.yml

```
create_env_script:
    class: File
    path: create_env.sh
    secondaryFiles:
      - class: File
        location: processing.json
      - class: Directory
        location: misfit_data

download_workflow:
    class: File
    path: download_FDSN.py
    secondaryFiles:
    - class: File
      location: download_helpers.py
    - class: File
      location: domain.py
    - class: File
      location: utils.py
download_argument_f:
    class: File
    path:  download_chile.json
preprocess_workflow:
  class: File
  path: create_misfit_prep.py
  secondaryFiles:
    - class: File
      location: preprocessing_functions.py
preprocess_argument_f:
    class: File
    path: misfit_input.jsn
ra_workflow:
  class: File
  path: dispel4py_RA.pgm_story.py
ra_argument_d: '{"streamProducerReal": [ {"input": "output/IV.ARRO.EHR.data"} ], "streamProducerSynth":
[ {"input": "output/IV.ARRO.HXR.synth"} ]}'
```

```
$ cwltool ra.cwl ra.yml
```

Available here